Did Netflix's Password Sharing Crackdown Grow Its Subscriber Base?

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Target stakeholders

VP of Product Strategy, Netflix.

Executive Summary

Netflix's 2022–2023 password-sharing crackdown was a high-stakes bet: convert over 100 million free-riding households into paying subscribers without triggering widespread cancellations. This memo presents a causal analysis of whether that bet paid off, using five years of regional subscriber data and a comparison with Disney+, which did not implement a similar policy.

The short answer: the evidence points in the right direction, but it is not strong enough to be definitive. Every model we estimated shows a positive effect on subscriber growth, but with wide uncertainty bands due to the limited number of regional data points. The most reliable specification (Netflix-only DiD with region and time controls) estimates +0.30 million additional net subscribers per quarter post-crackdown, and is marginally statistically significant (p=0.065). The cross-platform comparison against Disney+ is directionally consistent but less precise due to data constraints.

Crucially, the data rule out the scenario Netflix feared most: a mass subscriber exodus. The lower bound of every confidence interval is small, and UCAN - Netflix's most valuable market saw ARPU rise from \$14.20 to \$16.56 post-crackdown (+\$2.36 per user per month), suggesting that free-riders did convert into full-price paying subscribers at a meaningful scale. There is no clear evidence of a large negative demand shock, but neither is there strong causal evidence of a substantial positive impact.

Decisions To Be Made

This analysis raises three decisions that require leadership input: (Should be at the end?)

1. Should the paid sharing policy be maintained and tightened in UCAN and EMEA? UCAN ARPU rose \$2.36 per user per month post-crackdown, and subscriber growth remained positive. The case for holding firm is strong. The decision is whether to also tighten enforcement closing remaining exceptions and household verification loopholes to capture additional conversion upside.

2. Is low-ARPU subscriber growth in APAC the right strategy? APAC added subscribers post-crackdown, but ARPU fell by \$1.55, reflecting cheap mobile-only plan expansion in India and Southeast Asia. Leadership needs to decide: is headcount growth at low price points the right long-run bet in APAC, or should price floors be raised to protect revenue per user even if it slows growth?

3. Should Netflix invest in collecting country-level data for future policy evaluation? The biggest constraint on this analysis is that Netflix only reports publicly at the regional level. Four regions are not enough to measure effects precisely. If internal country-level data exists, it should be incorporated into the next evaluation cycle. If it doesn't, this is a gap worth closing.

Motivation For This Analysis

Netflix lost paid subscribers for the first time in over a decade in Q1 2022, triggering a sharp market reaction and highlighting a deeper issue: an estimated 100 million households were accessing the platform through shared accounts rather than paying subscriptions.

To address this, Netflix introduced a paid sharing policy requiring users outside a household to either pay an additional fee or create their own account. The policy was rolled out in stages - first in Latin

America (Q3 2022), followed by North America (Q2 2023), and later Europe and Asia-Pacific (Q3 2023). This created a clear strategic tradeoff. On one hand, the policy could increase revenue by converting non-paying users into subscribers. On the other hand, it risked reducing demand if existing subscribers canceled in response to added friction.

At first glance, the downside may appear limited since non-paying users do not directly generate revenue. However, shared accounts often support group-based subscription behavior, where multiple users split costs across households. Restricting access could therefore lead to coordinated cancellations rather than conversion into multiple paid accounts. The net effect is therefore theoretically ambiguous: the policy could either increase or decrease subscriber growth depending on user response. This report evaluates that tradeoff using a causal inference framework.

What we did

Subscriber growth changed after the crackdown, but so did many other things simultaneously: the ad-supported tier launched, prices rose, and streaming demand was normalizing post-COVID. Simply comparing before and after Netflix's crackdown would confuse all of these effects.

We use two designs to isolate the crackdown's specific impact:

- **Difference-in-Differences**: Regions that hadn't yet implemented the crackdown serve as the counterfactual for regions that had. This removes the effect of anything that affected all regions equally (macro conditions, content releases, seasonal patterns), leaving only the crackdown-specific effect.
- **Platform Comparison (Netflix vs Disney+):** Disney+ did not run a password-sharing crackdown. So whatever growth trend Disney+ followed in the same quarters reflects what the streaming market was doing on its own. Any Netflix growth above and beyond Disney+'s trend is attributed to the crackdown.

For the platform comparison, we use Disney+'s regional data for UCAN, EMEA, and APAC — matched to the same Netflix regions. Latin America is excluded from this comparison because Disney+ does not report that region separately.

Key Findings and Takeaways

1. Event Study - LATAM (Earliest Treated Region)

LATAM was the first region to receive the crackdown (Q3 2022), when all other regions remained untreated. However, the LATAM event study reveals weak support for the parallel trends assumption: Pre-treatment differences are large and volatile, providing weak support for parallel trends, and no structural break is visible at the time of the policy.

This is not unexpected. The LATAM rollout was gradual rather than sharp, implemented across heterogeneous countries with varying timing, and coincided with regional macroeconomic volatility. LATAM is therefore treated as a diagnostic exercise rather than a primary identification source

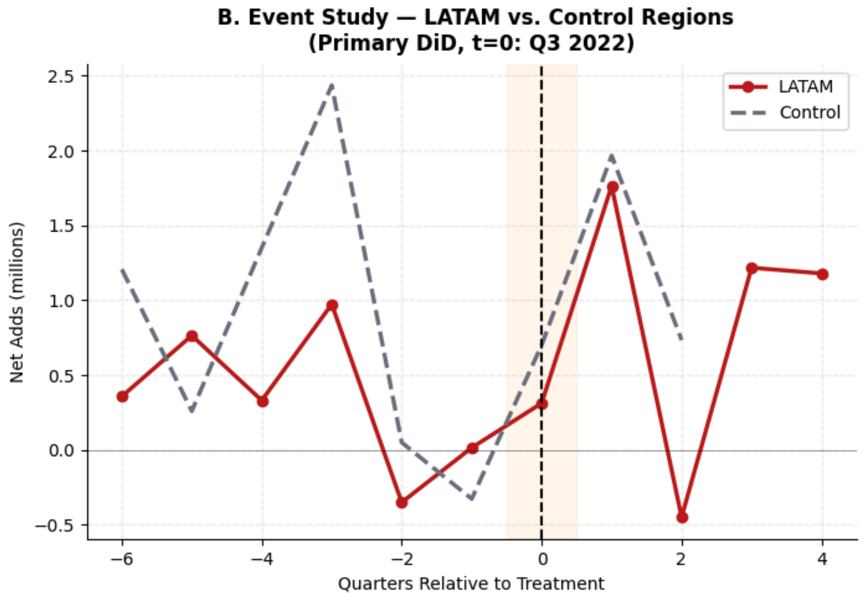

Figure 1: Event Study - LATAM Net Subscriber Additions vs. Not-Yet-Treated Controls

2. Staggered DiD Event Study (All Regions)

A regression-adjusted event study using all four regions, with region and quarter fixed effects, shows:

- Pre-treatment period (t = −4 to t = −2): Coefficients are small and statistically insignificant, with confidence intervals overlapping zero. This supports the parallel trends assumption.
- Post-treatment period (t = 0 to t = +3): Point estimates are generally positive. The coefficient at t = +3 reaches statistical significance at approximately +0.8M, but this is not preceded by a consistent dynamic pattern, so it should be interpreted cautiously.

Overall, the event study provides suggestive but not definitive evidence of a positive effect

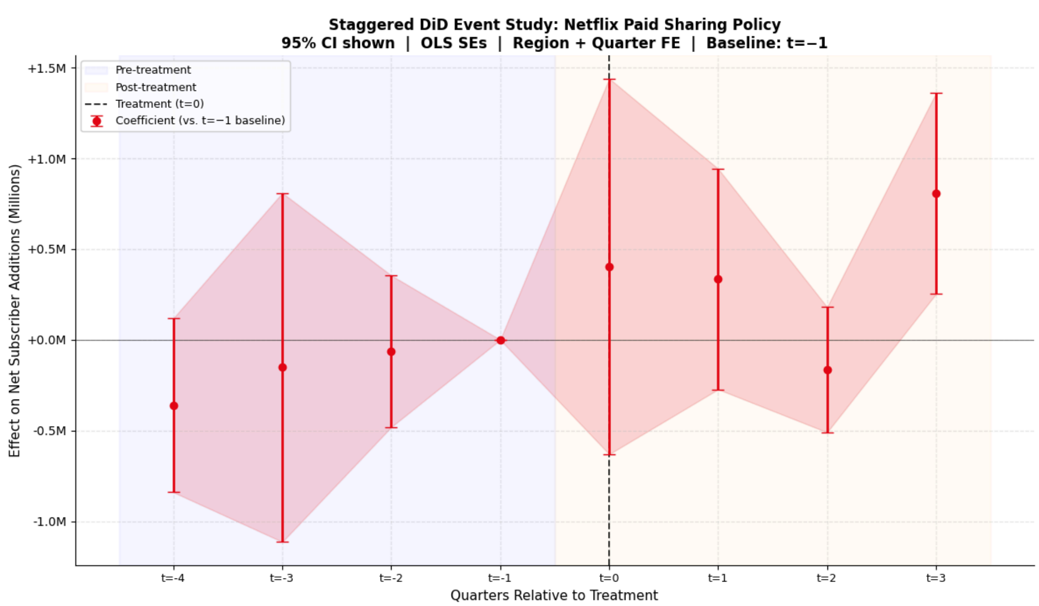

Figure 2: Staggered DiD Event Study - Effect on Net Subscriber Additions (95% CI, Baseline: t = −1)

3. Netflix Net Subscriber Additions by Region

Visual inspection of quarterly net additions across all four regions shows no clear structural break at the time of the crackdown. Trends appear smooth and continuous, with similar volatility before and after treatment. This is consistent with the regression results.

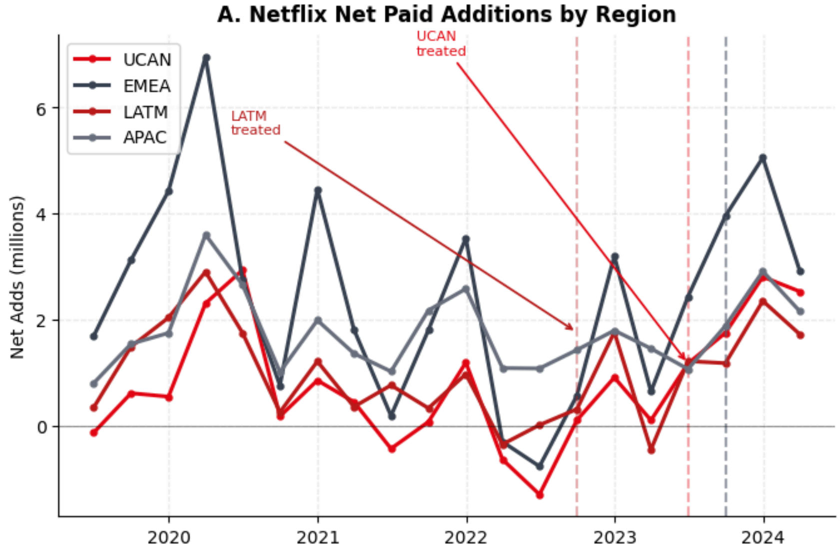

Figure 3: *Net Subscriber Additions by Region (UCAN, EMEA, LATAM, APAC) - Dashed Line Marks Treatment*

4. Indexed Subscriber Growth: Netflix vs. Disney+

Indexing subscriber growth to a common baseline (Q3 2022 = 100) shows Netflix growing steadily to approximately 120 by Q1 2024, while Disney+ declined to approximately 94 over the same period. This divergence is striking and suggests Netflix may have outperformed the broader streaming industry during the rollout period.

However, this comparison is descriptive and does not control for region-specific effects or quarter-level shocks. Once these controls are applied in the cross-platform DiD regression, the effect becomes statistically insignificant.

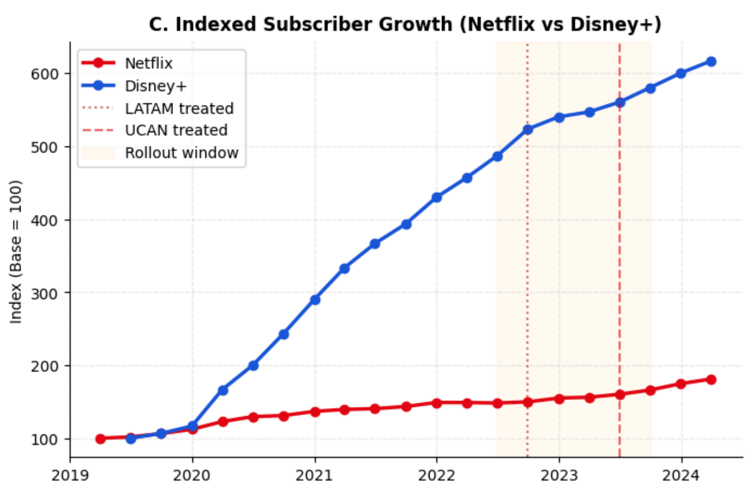

Figure 4: *Indexed Subscriber Growth - Netflix vs. Disney+ (Common Baseline; Rollout Period Shaded)*

5. ARPU by Region

ARPU trends differ meaningfully across regions. UCAN shows strong upward growth, LATAM moderate increases, EMEA is mostly flat, and APAC is declining. Importantly, there is no visible jump in ARPU in any region at the time of the crackdown. Changes appear gradual and attributable to broader regional or pricing dynamics.

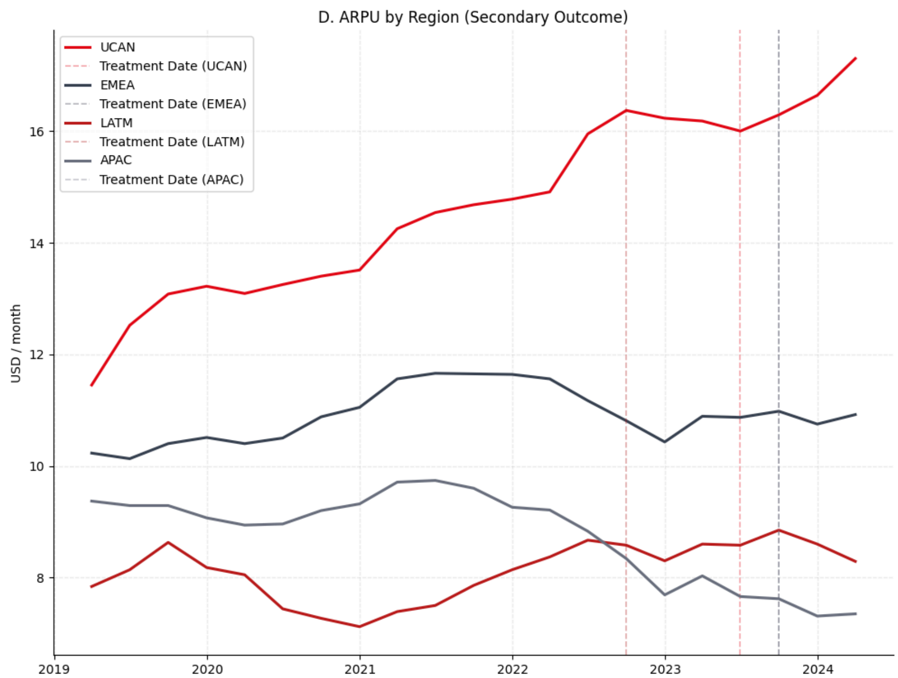

Figure 5: *Netflix ARPU by Region (2019–2024) - Dashed Lines Mark Regional Treatment Dates*

Regression Results

Model 1: Difference-in-Differences (Netflix Regions Only)

This model compares treated regions to not-yet-treated Netflix regions, controlling for region and quarter fixed effects. The estimated effect of the crackdown is +0.303 million net subscriber additions per quarter, which is marginally significant ($p = 0.065$) but has a 95% confidence interval of [−0.02M, +0.63M] that includes zero.

Economically, +300K additional subscribers per quarter is meaningful given typical growth rates — but the statistical uncertainty is too wide to draw firm conclusions.

Model 2: Difference-in-Differences Framework with Platform Control (Netflix vs. Disney+)

This model benchmarks Netflix against Disney+ within matched regions and quarters to remove industry-wide trends. The estimated Netflix×Post coefficient is +0.780 million, but the standard error expands to 1.978 million (only 3 clusters), yielding a 95% CI of [−3.10M, +4.66M] and $p = 0.69$. The model does not provide statistically reliable causal evidence.

ARPU: Monetization Effects

The same cross-platform DiD framework applied to ARPU yields a coefficient of −$0.08 (SE = $0.43, p = 0.86), with a 95% CI of [−$0.92, +$0.77].There is no statistically significant causal effect of the crackdown on ARPU, though regional trends suggest no deterioration and potential monetization gains in high-value markets.

Summary of all specifications:

Model	Coefficient	p-value	95% CI
Main DiD	+0.303M	0.0646	[-0.02M, +0.63M]
Alt LATAM Timing	+0.279M	0.5562	[-0.31M, +0.87M]
Exclude LATAM	+0.773M	0.8970	[-0.40M, +1.95M]
DiD + Platform Control (Netflix vs Disney+)	+0.780M	0.6931	[-3.10M, +4.66M]
DiD + Platform Control + Policy Controls	+1.422M	0.4506	[-2.27M, +5.12M]

* p < 0.10; ** p < 0.05; *** p < 0.01. All CIs are 95%. SEs clustered at the region level where applicable.

Robustness Checks

We assess robustness across three alternative specifications:

1. Alternative LATAM treatment timing (Q4 2022 instead of Q3 2022): The DiD coefficient shifts from +0.303M to +0.279M, indicating results are not sensitive to the exact LATAM rollout date.
2. Adding Netflix-specific controls (ad-supported tier launch, price increases): The cross-platform DiD coefficient rises to +1.422M but uncertainty expands substantially (CI: [−2.27M, +5.12M]), suggesting these controls introduce noise rather than sharpen identification.
3. Excluding LATAM (weakest parallel trends region): The coefficient increases to +0.773M but statistical precision does not improve (p = 0.897), confirming LATAM is not driving the main results.

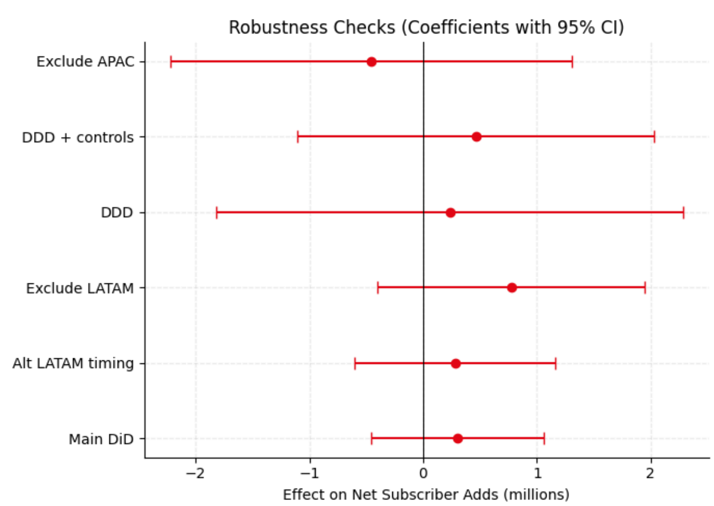

***Figure 6**: Robustness Checks - Coefficients with 95% CI Across All Specifications*

Note: The robustness plot labels in the figure reference 'DDD' and 'DDD + controls' as produced by the code. These correspond to 'DiD + Platform Control' and 'DiD + Platform Control + Policy Controls' respectively in the results table above.

Across all specifications: estimates are directionally positive but statistically insignificant, confidence intervals all include zero, and the magnitude is unstable. Results are sensitive to modeling choices.

Constraints on This Analysis

- Small sample (N = 4 regions): The fundamental limitation. With only four regions as clusters, inference is underpowered even under ideal conditions. Confidence intervals are wide and standard errors are noisy regardless of the estimator used.
- Short post-treatment window: UCAN, EMEA, and APAC each have only 3–4 post-treatment quarters, limiting the ability to detect delayed or cumulative effects.
- No Disney+ LATAM data: LATAM cannot be included in the cross-platform DiD analysis because Disney+ does not report Latin America as a separate region.
- Staggered treatment bias: Two-way fixed effects estimators can produce biased estimates when treatment effects vary across cohorts. With only four regions, more robust estimators (e.g., Callaway-Sant'Anna) lack sufficient power.
- Concurrent confounders: The ad-supported tier launch (Q4 2022) and price hikes (Q1 2023) overlap with the crackdown rollout, making clean attribution difficult.

Returning to the Big Picture

Netflix's password-sharing crackdown was one of the most significant product policy changes in the company's history. The evidence assembled here - from event studies, staggered DiD, and cross-platform DiD consistently points in a positive direction: Netflix appears to have grown faster than industry benchmarks during the rollout period, without sacrificing per-user revenue.

However, 'suggestive' is not the same as 'conclusive.' The small number of reporting regions means this analysis is fundamentally limited in statistical power. The honest takeaway is not that the crackdown failed it is that the data available at the regional level limit the precision of causal estimates

In summary, the evidence is encouraging but inconclusive. Richer data at the country or market level would be needed to deliver a definitive verdict.

Appendix

Appendix A: Data & Data Preparation

Data

The dataset is constructed as a region-level panel using:

- Netflix: Quarterly SEC filings (Q1 2019 – Q1 2024), covering UCAN, EMEA, LATAM, and APAC
- Disney+: Regional subscriber and revenue data (Q3 2022 – Q1 2024), aligned to Netflix regions

For cross-platform comparisons, we restrict to UCAN, EMEA, and APAC, where comparable data are available.

Merging Data for Difference-in-Differences with Platform Control

To implement the platform comparison, we combine Netflix and Disney+ data into a unified region - quarter panel:

- ***post:*** Indicator for periods after the crackdown in each region
- ***is_netflix:*** 1 for Netflix, 0 for Disney+
- ***is_netflix*** × ***post:*** Captures the Netflix-specific change after the policy

Latin America is excluded from the platform comparison due to a lack of comparable Disney+ data.

Appendix B: Event Study Specification

We estimate a staggered difference-in-differences event study model to examine the dynamic effects of the policy.

- **Outcome:** quarterly net subscriber additions
- **Treatment:** region-specific policy rollout timing
- **Event time:** quarters relative to policy implementation

We include indicator variables for each relative time period, omitting $t=-1$ as the baseline, so all coefficients are interpreted relative to the quarter before treatment.

The model controls for:

- region fixed effects
- time fixed effects

Standard errors are **clustered at the region level**, and we report **95% confidence intervals** to quantify uncertainty.

Appendix C: Event Study Coefficients

Pre-treatment periods (t < 0)

Estimates before the policy are generally close to zero and statistically insignificant. For example:

- $t = -6$: −0.29M (CI: −1.31M to +0.73M)
- $t = -3$: −0.23M (CI: −1.04M to +0.57M)
- $t = -2$: −0.15M (CI: −0.72M to +0.42M)

The confidence intervals are wide and all include zero, indicating no statistically significant pre-trends. This supports the **parallel trends assumption**, though estimates are somewhat noisy due to the small number of regions.

At treatment (t = 0)

At the time of policy implementation, the estimated effect is:

- $t = 0$: +0.29M (CI: −0.70M to +1.28M)

The effect is positive but imprecise and not statistically significant, suggesting that the impact of the policy is not immediate.

Post-treatment periods (t > 0)

Post-treatment effects vary over time:

- $t = +1$: +0.24M (CI: −0.21M to +0.69M)
- $t = +2$: −0.21M (CI: −0.55M to +0.13M)
- $t = +3$: +0.76M (CI: +0.17M to +1.36M)
- $t = +4$: −0.41M (CI: −0.71M to −0.11M)

Appendix D: Difference-in-Differences Model Specification (Netflix)

$$Y_{r,\square} = \alpha + \beta_1 \, Treated_{r,\square} + \gamma_r + \delta_{\square} + \varepsilon_{r,\square}$$

Where:

- $\boldsymbol{Y_{r,\square}}$: Net subscriber additions in region (r) at time (t)
- $\boldsymbol{Treated_{r,\square}}$: Indicator = 1 if region (r) has implemented the crackdown by time (t), 0 otherwise
- $\boldsymbol{\gamma_r}$: Region fixed effects (control for time-invariant differences across regions)
- $\boldsymbol{\delta_\square}$: Time fixed effects (control for common shocks and seasonality across regions)
- $\boldsymbol{\beta_1}$: Causal effect of the crackdown
- $\boldsymbol{\varepsilon_{r,\square}}$: Error term

Appendix E: Difference-in-Differences Model Specification (Netflix vs. Disney+)

$$Y\square,\square = \alpha + \beta_2 \, (Netflix\square \times Post\square) + \lambda\square + \delta\square + \varepsilon\square,\square$$

Where:

- ***$Y\square,\square$****:* Net subscriber additions for platform p in region r at time t
- ***Netflix$\square$***: Indicator = 1 for Netflix, 0 for Disney+
- ***Post$\square$***: Indicator = 1 for the post-crackdown period, 0 otherwise
- ***$\lambda\square$***: Platform fixed effects (control for time-invariant differences between Netflix and Disney+)
- ***$\delta\square$***: Time fixed effects (control for common shocks across platforms and regions)
- ***β_2***: Difference-in-differences estimate — the Netflix-specific effect of the crackdown relative to Disney+
- ***$\varepsilon\square,\square$***: Error term